

13. Standardmodell der Teilchenphysik

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- 3 Familien von Fermionen

$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}_L \quad \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}_L \quad \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}_L$$

$$e_R^-, \mu_R^-, \tau_R^-$$

$$\begin{pmatrix} u \\ d' \end{pmatrix}_L \quad \begin{pmatrix} c \\ s' \end{pmatrix}_L \quad \begin{pmatrix} t \\ b' \end{pmatrix}_L$$

$$u_R, c_R, t_R \\ d_R, s_R, b_R$$

- elektroschwache und starke WW

$$W^\pm, Z^0, \gamma: \text{erhaltene Ladungen } T, T_3, Y \rightarrow Q = T_3 - \frac{Y}{2}$$

$$SU(2)_L \otimes U(1)_Y \quad U(1)_{em}$$

$$\text{Glueon: erhaltene Ladung } r, g, b$$

$$SU(3)$$

$\Rightarrow$ Symmetrie des SM

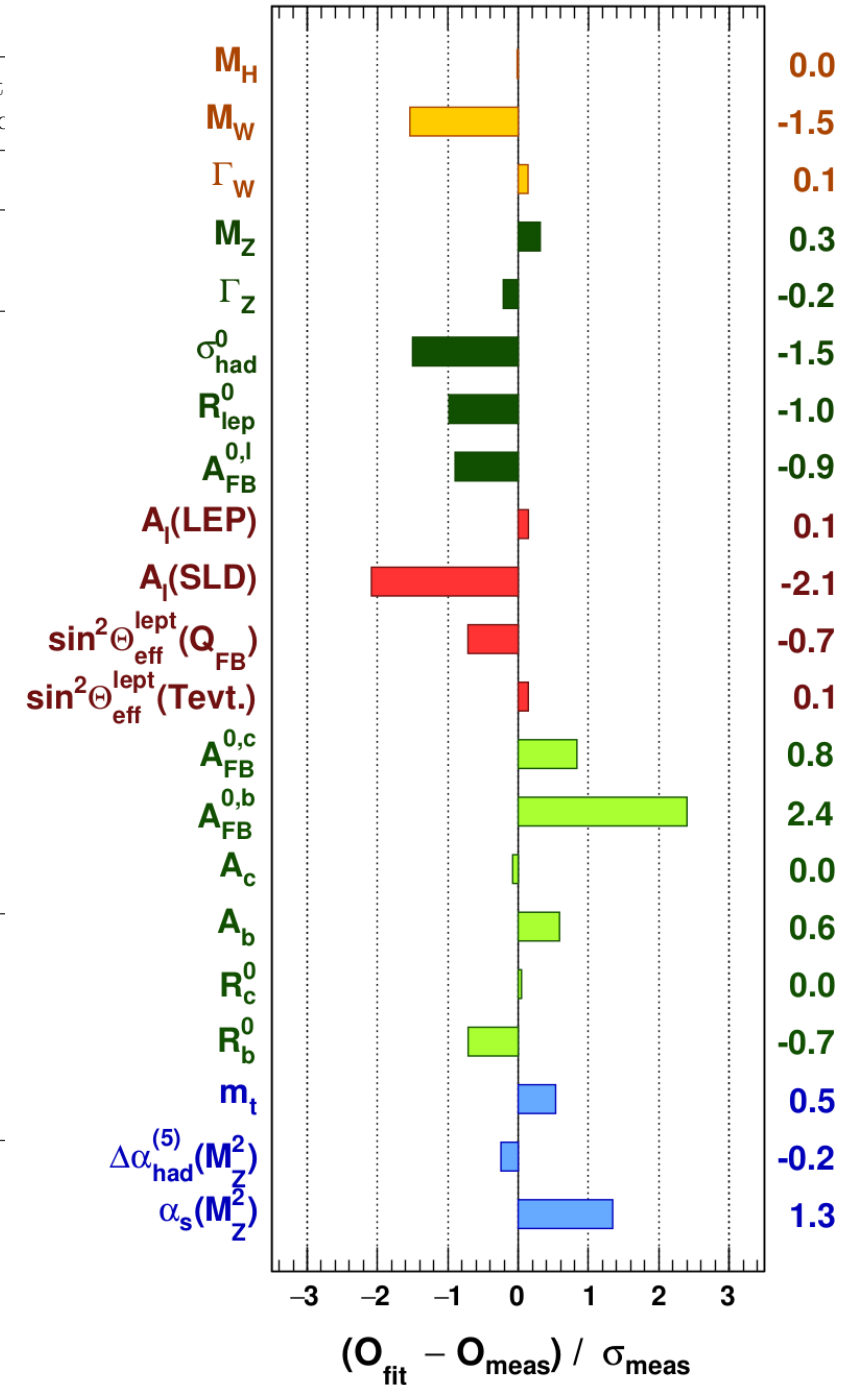
$$SU(3)_c \otimes SU(2)_L \otimes U(1)_Y$$

- Higgs

Präzisionstests $\rightarrow$ SM hervorragend bestätigt

arXiv:1803.01853

Parameter	Input value	Free in fit	Fit Result	Fit w/o exp. input in line	Fit w/o exp. input in line, no theo. unc
M_H [GeV]	125.1 ± 0.2	yes	125.1 ± 0.2	90^{+21}_{-18}	89^{+20}_{-17}
M_W [GeV]	80.379 ± 0.013	–	80.359 ± 0.006	80.354 ± 0.007	80.354 ± 0.005
Γ_W [GeV]	2.085 ± 0.042	–	2.091 ± 0.001	2.091 ± 0.001	2.091 ± 0.001
M_Z [GeV]	91.1875 ± 0.0021	yes	91.1882 ± 0.0020	91.2013 ± 0.0095	91.2017 ± 0.0089
Γ_Z [GeV]	2.4952 ± 0.0023	–	2.4947 ± 0.0014	2.4941 ± 0.0016	2.4940 ± 0.0016
σ_{had}^0 [nb]	41.540 ± 0.037	–	41.484 ± 0.015	41.475 ± 0.016	41.475 ± 0.015
R_{ℓ}^0	20.767 ± 0.025	–	20.742 ± 0.017	20.721 ± 0.026	20.719 ± 0.025
$A_{\text{FB}}^{0,\ell}$	0.0171 ± 0.0010	–	0.01620 ± 0.0001	0.01619 ± 0.0001	0.01619 ± 0.0001
$A_{\ell}^{(*)}$	0.1499 ± 0.0018	–	0.1470 ± 0.0005	0.1470 ± 0.0005	0.1469 ± 0.0003
$\sin^2 \theta_{\text{eff}}^{\ell}(Q_{\text{FB}})$	0.2324 ± 0.0012	–	0.23153 ± 0.00006	0.23153 ± 0.00006	0.23153 ± 0.00004
$\sin^2 \theta_{\text{eff}}^{\ell}(\text{TeVt.})$	0.23148 ± 0.00033	–	0.23153 ± 0.00006	0.23153 ± 0.00006	0.23153 ± 0.00004
A_c	0.670 ± 0.027	–	0.6679 ± 0.00021	0.6679 ± 0.00021	0.6679 ± 0.00014
A_b	0.923 ± 0.020	–	0.93475 ± 0.00004	0.93475 ± 0.00004	0.93475 ± 0.00002
$A_{\text{FB}}^{0,c}$	0.0707 ± 0.0035	–	0.0736 ± 0.0003	0.0736 ± 0.0003	0.0736 ± 0.0002
$A_{\text{FB}}^{0,b}$	0.0992 ± 0.0016	–	0.1030 ± 0.0003	0.1032 ± 0.0003	0.1031 ± 0.0002
R_c^0	0.1721 ± 0.0030	–	0.17224 ± 0.00008	0.17224 ± 0.00008	0.17224 ± 0.00006
R_b^0	0.21629 ± 0.00066	–	0.21582 ± 0.00011	0.21581 ± 0.00011	0.21581 ± 0.00004
$\bar{m}_c$ [GeV]	$1.27^{+0.07}_{-0.11}$	yes	$1.27^{+0.07}_{-0.11}$	–	–
$\bar{m}_b$ [GeV]	$4.20^{+0.17}_{-0.07}$	yes	$4.20^{+0.17}_{-0.07}$	–	–
m_t [GeV] ^(∇)	172.47 ± 0.68	yes	172.83 ± 0.65	176.4 ± 2.1	176.4 ± 2.0
$\Delta\alpha_{\text{had}}^{(5)}(M_Z^2)^{(\dagger\Delta)}$	2760 ± 9	yes	2758 ± 9	2716 ± 39	2715 ± 37
$\alpha_s(M_Z^2)$	–	yes	0.1194 ± 0.0029	0.1194 ± 0.0029	0.1194 ± 0.0028

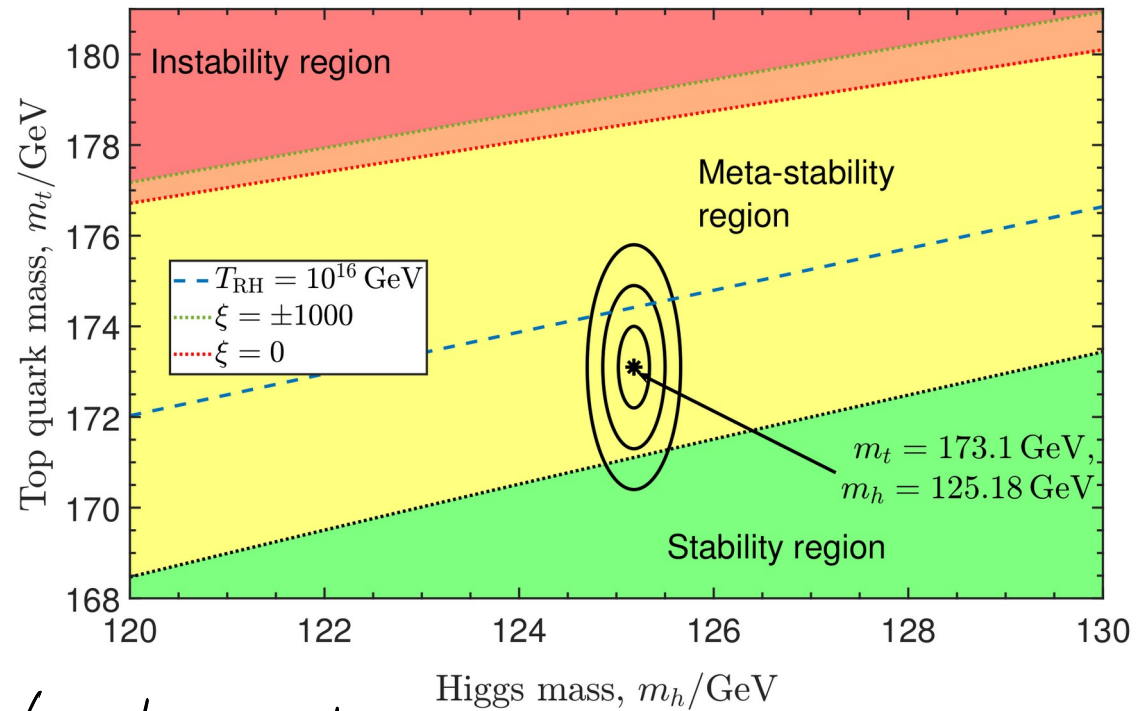
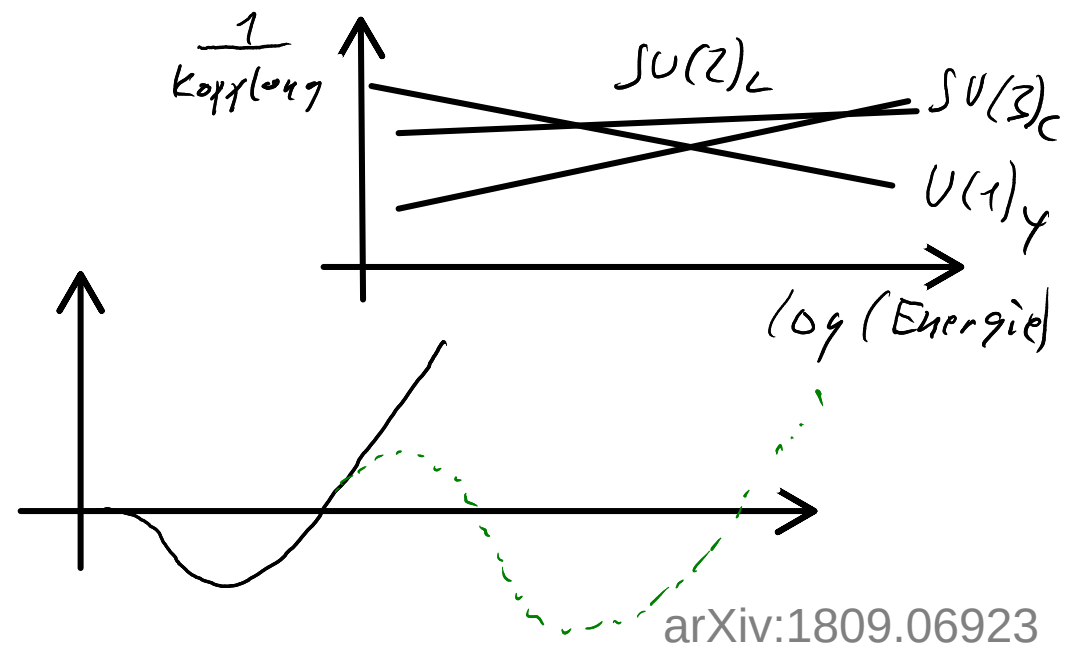


13.1 Offene Fragen

- Materie - Antimaterie - Asymmetrie im Universum
- Dunkle Materie (+ Energie)
- Neutrinomasse
- Parameter (strong CP problem)
- Hierarchieproblem
 $H \cdots \bigcirc \cdots$
- Vakuumstabilität
- Vereinheitlichung (GUT)
- Gravitation

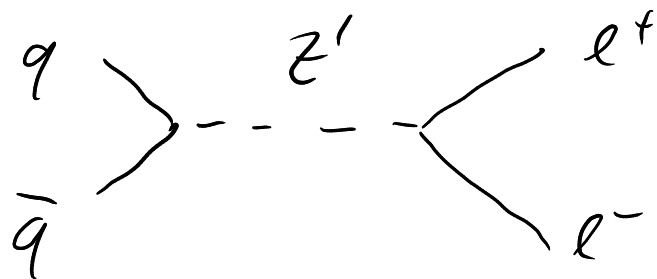
Vieffältige Erklärungsansätze:

Supersymmetrie, Stringtheorie, Leptoquarks, Axionen,

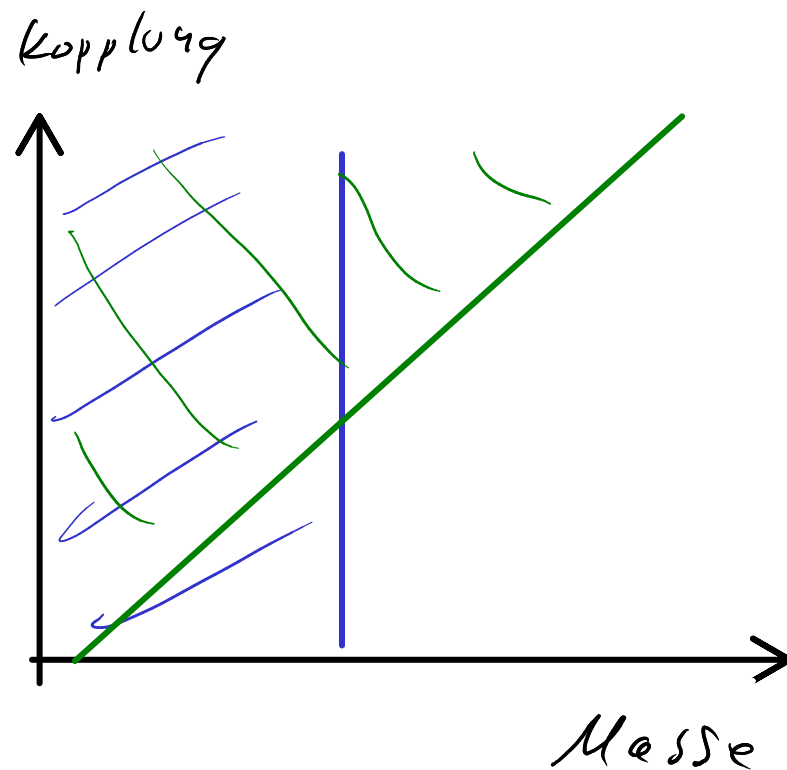


Experimentelle Suche nach neuer Physik

direkt: Erzeugung neuer Teilchen in
hochenergetischen Kollisionen



indirekt: Nachweis von Beiträgen
virtueller neuer Teilchen

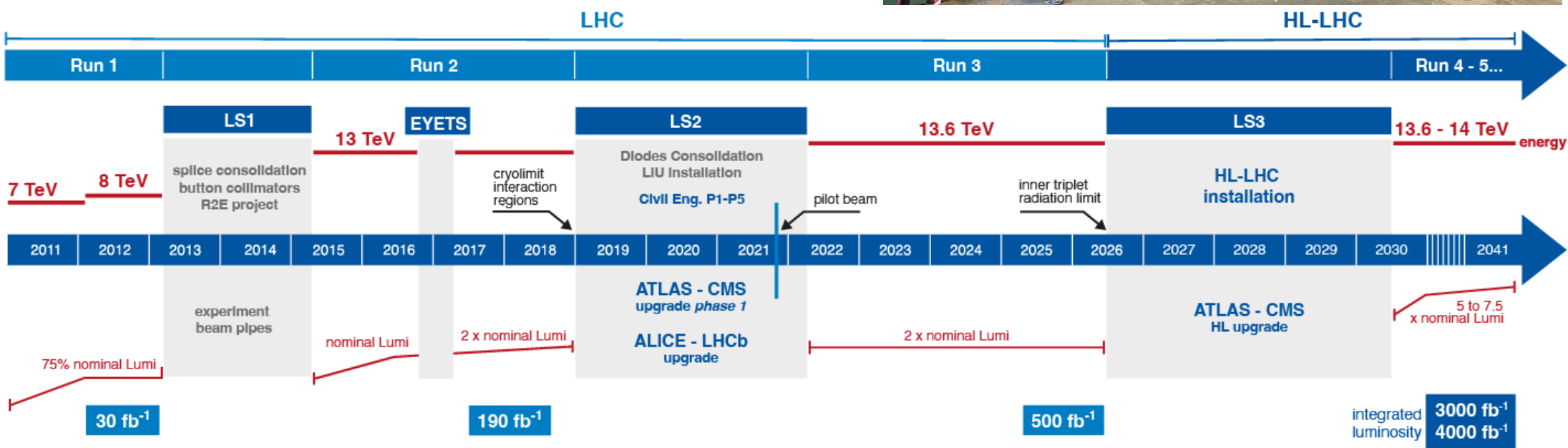


High-Lumi - LHC

Upgrades von ATLAS, CMS,
LHCb



LHC / HL-LHC Plan



HL-LHC TECHNICAL EQUIPMENT:

DESIGN STUDY



PROTOTYPES

CONSTRUCTION

INSTALLATION & COMM.

PHYSICS

HL-LHC CIVIL ENGINEERING:

DEFINITION

EXCAVATION

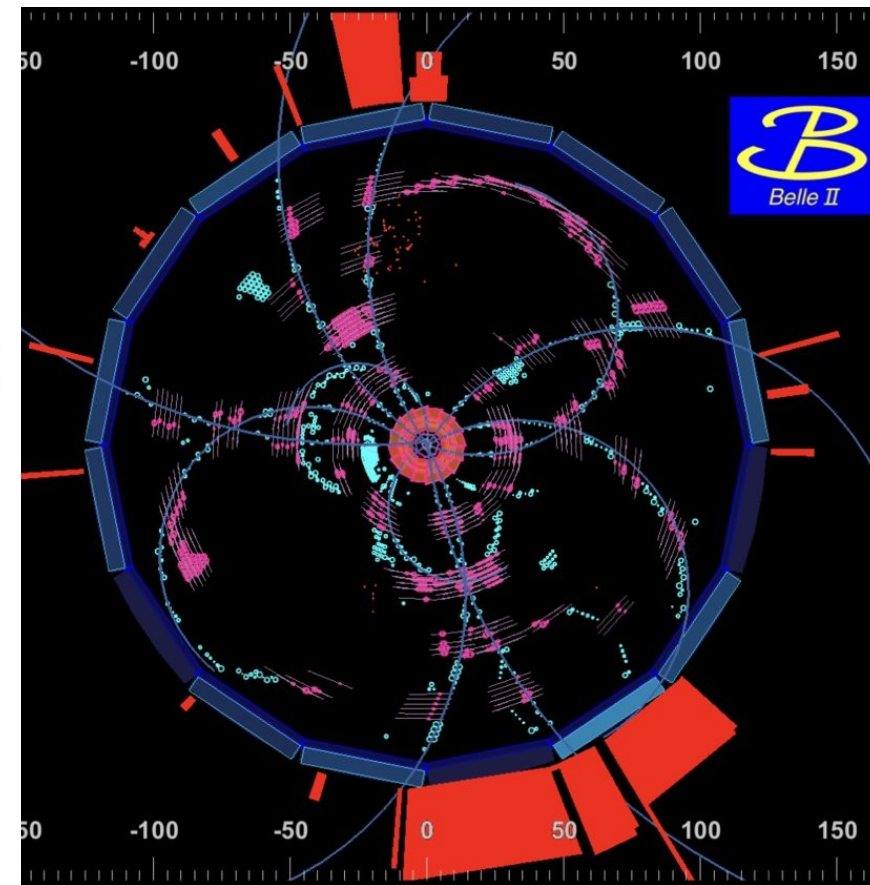
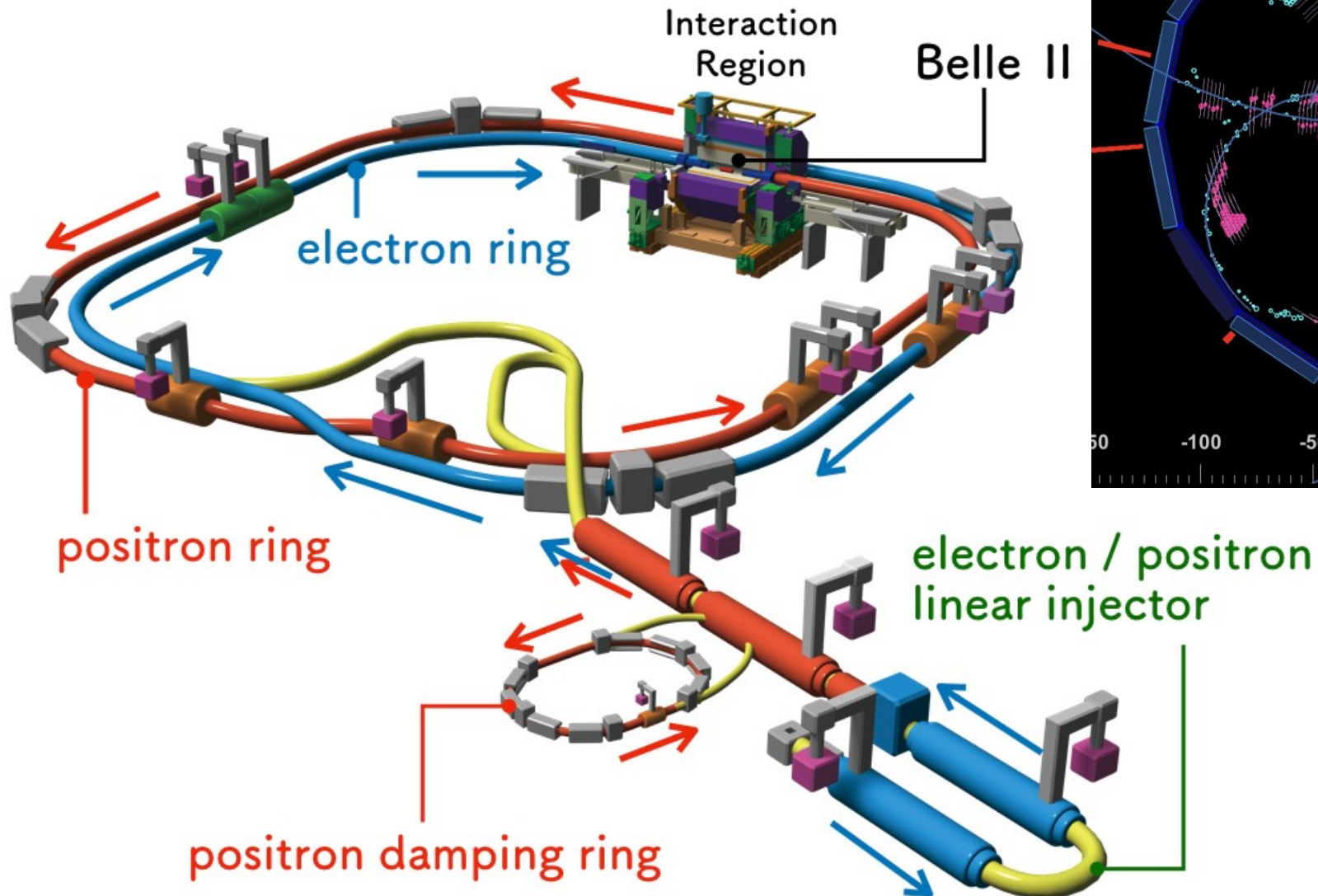
BUILDINGS

CORES

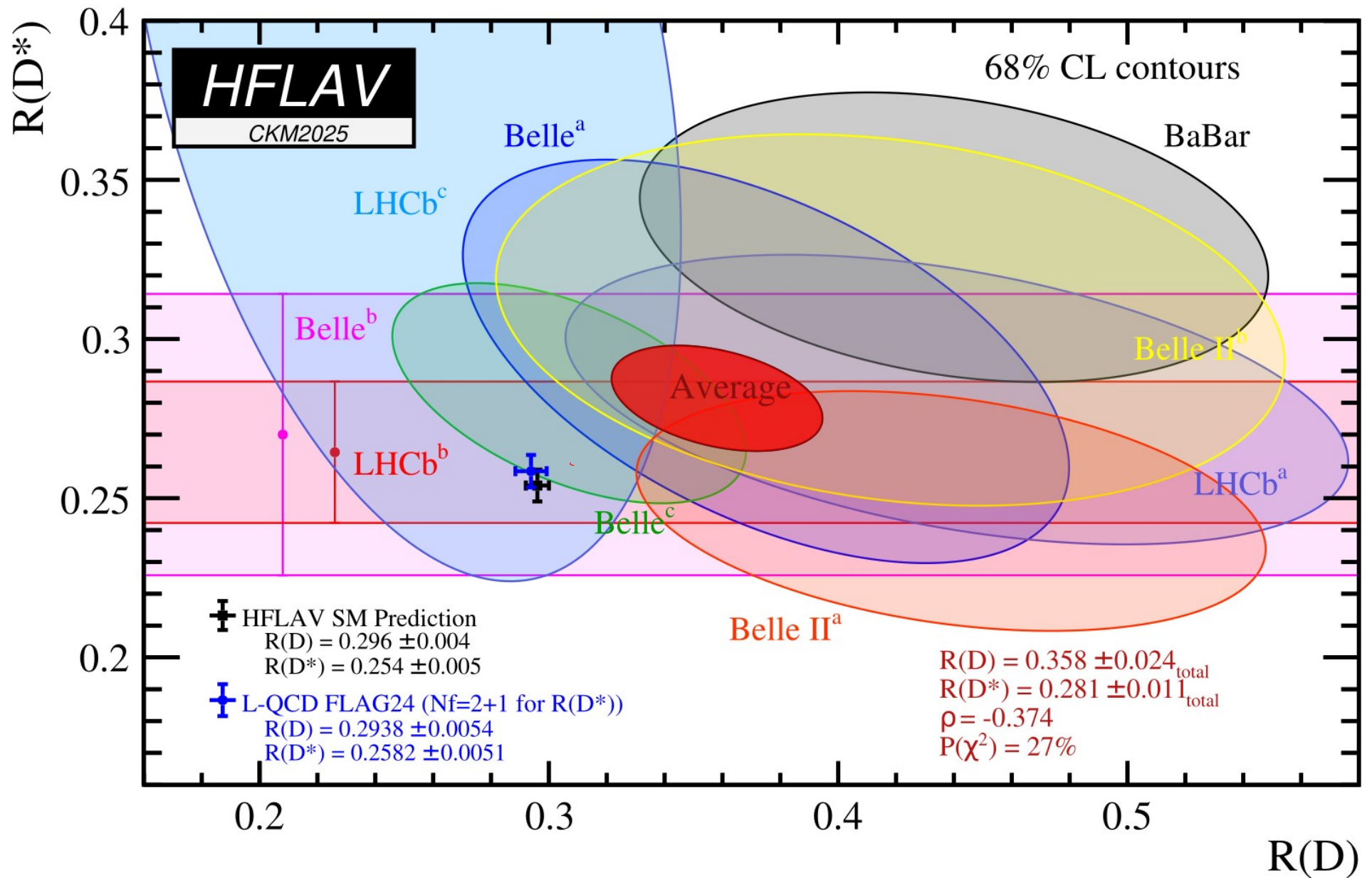
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SuperKEKB, Belle II

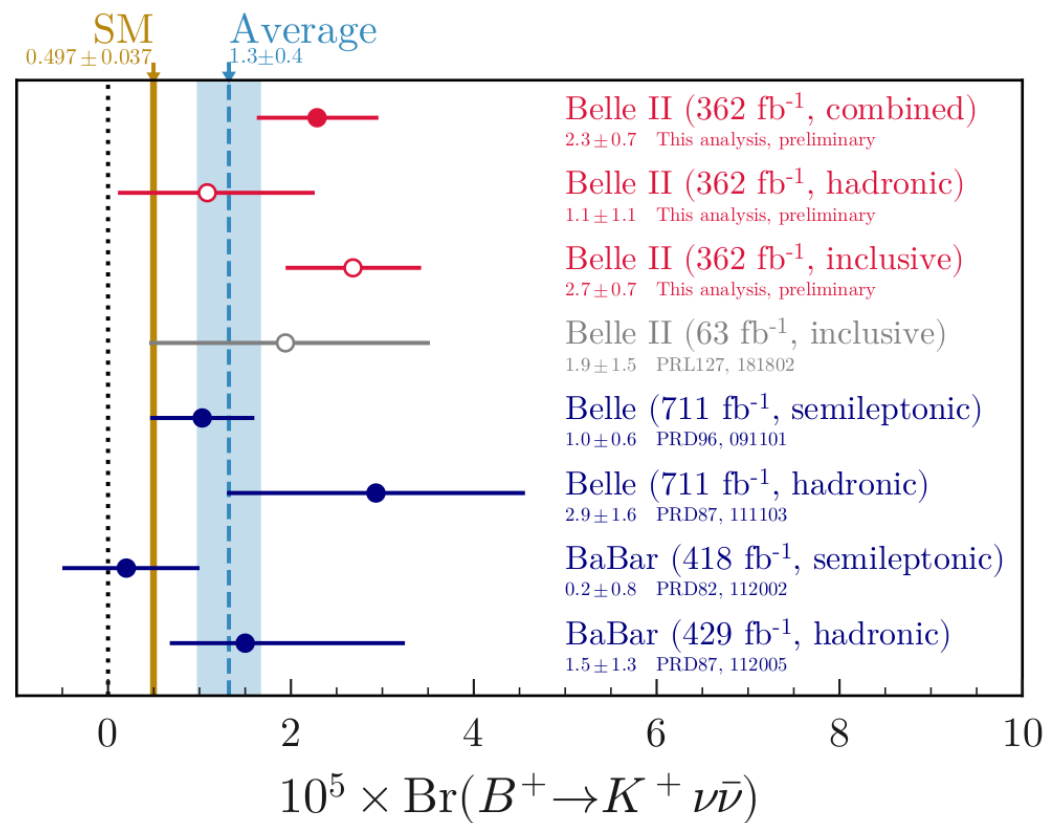
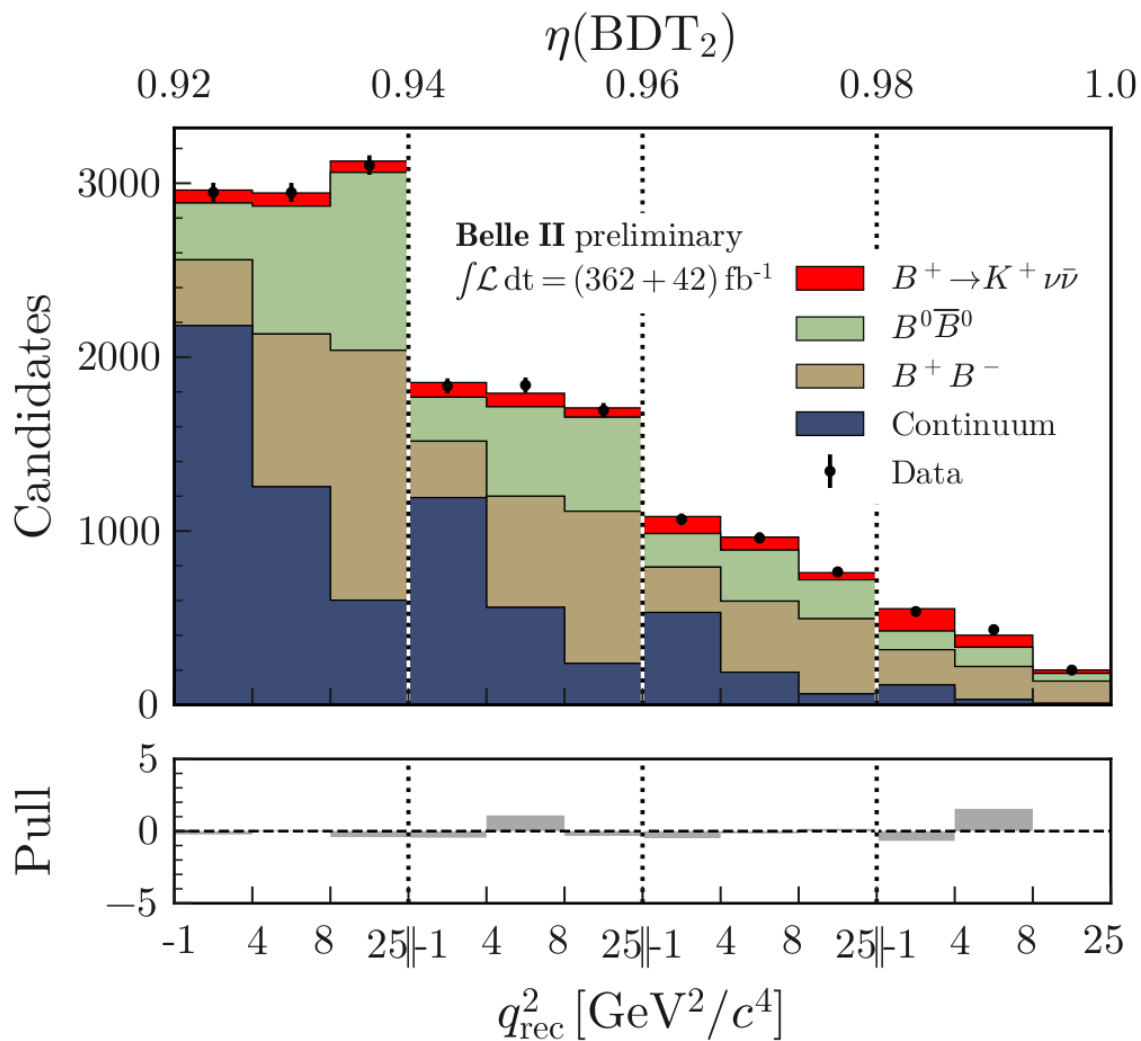
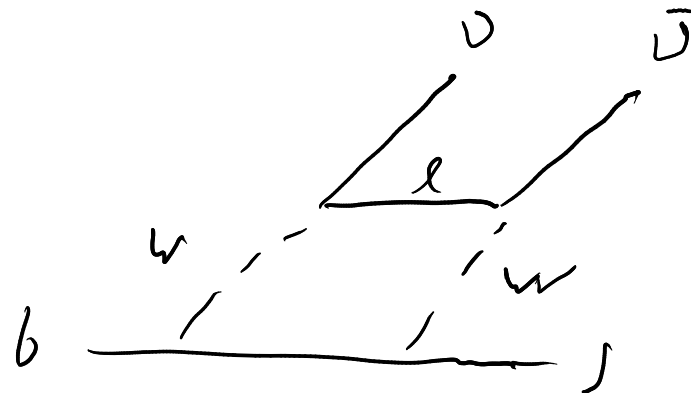
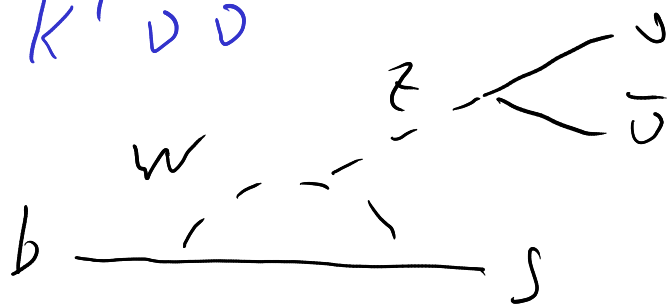
Ziel: 50 x mehr Daten



$$R(D^{(*)}) = \frac{BR(B \rightarrow D^{(*)} e^- \bar{\nu}_e)}{BR(B \rightarrow D^{(*)} \mu^- \bar{\nu}_\mu)}, \quad \ell = e, \mu$$



$$B^+ \rightarrow K^+ \nu \bar{\nu}$$





FUTURE CIRCULAR COLLIDER

